

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

B.Tech (Electronics and Telecommunication Engineering) SEMESTER - 4 Summer 2025 (Regular)

Course : Bachelor of Technology (Electronics and Telecommunication Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 4

Subject Code & Name: BTBS404 - PROBABILITY THEORY AND RANDOM PROCESSES

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

1 A sample space is:

- A. A set of all favorable outcomes
- B. A set of all possible outcomes of an experiment
- C. A subset of events
- D. The number of ways an event can occur

An event that contains no outcomes is called:

- A. Sample space
- B. Sure event
- C. Independent event
- D. Null event

3 Two events A and B are mutually exclusive if:

- A. $P(A \cap B) = 0$
- B. $P(A | B) = P(A)$
- C. $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- D. $P(B | A) = 1$

A random variable is defined as:

- A. A function mapping sample space to outcomes
- B. An outcome of a random process
- C. A function that assigns a real number to each outcome in the sample space
- D. A deterministic value

5 Which of the following is a discrete random variable?

- A. Height of students
- B. Number of heads in 3 coin tosses
- C. Temperature
- D. Speed of a moving car

6 The area under the PDF curve between two points a and b gives:

- A. Expectation
- B. Mean value
- C. $P(a \leq X \leq b)$
- D. Variance

7 If $Z=aX+b$, then the mean of Z is:

- A. $aE[X]+b$ B. $aE[X]$
- C. $a+b$ D. $E[X]+ab$

8 A random vector is defined as:

- A. A scalar-valued function of a random variable
- B. A deterministic vector
- C. A vector whose elements are random variables
- D. A probability function

The covariance matrix of a random vector is always:

- A. Symmetric and positive definite (if non-degenerate)
- B. Skew-symmetric
- C. Diagonal only
- D. Singular

10 A binomial distribution models:

- A. The number of failures before success
- B. The number of successes in a fixed number of independent Bernoulli trials
- C. Continuous outcomes between 0 and 1
- D. The time until the next event

A random process is defined as:

- A. A single deterministic signal over time
- B. A collection of random variables indexed by time or space
- C. A function of probability
- D. A distribution of values

A wide-sense stationary (WSS) process requires:

- A. Mean to be constant and autocorrelation depends only on time difference
- B. Zero mean and zero variance
- C. Only variance is time-invariant
- D. First and third moments are constant

Q2. Solve the following.

Define probability with its all approaches (numerical & classical) approaches 6

Explain disadvantages of classical approach of probability

Explain the probability density function (PDF) and the cumulative distribution function (CDF) along with their properties for continuous and discrete random variables. 6

Q3. Solve the following.

A) If X is uniformly distributed in $(1,2)$ and $Y=X^3$, then find the mean and variance of Y . 6

B) State and prove the statement of the Schwarz Inequality. 6

Q4. Solve Any Two of the following.

A) State and explain the Poisson distribution. Also, write the mean and variance formula for the Poisson distribution. 6

- B) If 3% of electronic units manufactured by a company are defective. Find the probability that in a sample of 200 units, less than 2 bulbs are defective. 6
- C) State and explain the terms Covariance and Correlation and write the properties. 6

Q5. Solve Any Two of the following.

- A) Write a note on the law of large numbers and explain the types. 6
- B) State and prove the central limit theorem. 6
- C) Explain in detail axioms of probability. Find the value of k 6

x	1	2	3	4
p(x)	0.1	0.4	0.1	k

Find 1) $p(x < 3)$ 2) $p(2 < x = 4)$

Q6. Solve Any Two of the following.

- A) If $\{X(t)\}$ is WSS process with autocorrelation $R(\tau) = Ae^{-a|\tau|}$. Determine the second-order moment of RV $X(9) - X(4)$. 6
- B) Define the power spectral density function. Write all its properties and prove any three. 6
- C) Define random process. Provide the classification of random process based on nature of space and time parameter along with example for each type. 6

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